of users only retry a mobile app once or at the most, twice, if it fails to work as intended or if it is slow. It is important to keep users engaged at all times and that can be done via fast performing native apps. One key example is of Facebook migrating from HTML to a native mobile app to keep users engaged by serving more feeds at the same time.

   b. If it is going to be an in-house app for users within the organisation, then hybrid apps can work, particularly if the app is meant for routine business operations. Internal apps need not be very exciting for users, as long as they solve the purpose in the expected timeframe.

3. What is the geography of your majority user base? How many platforms do you actually want to cover?
   a. If you are targeting just one geographic region with the launch of your app, then native is a better choice, as you can just target the one platform that commands the highest market share in that geographic region (like iOS for USA).
   b. If you are targeting to release your app globally, then a hybrid app can be opted for as the market will be shared by different mobile OSs/platforms.
4. How big or rich does your mobile application need to be?
   a. If you are targeting to launch a full blown, rich mobile app with built-in device hardware capabilities like a camera, GPS, etc, then native is a better choice for enhancing the performance of the app, and effectively achieving touch/gesture handling and built-in device hardware capabilities.
   b. If you are targeting to launch a minimum viable product (MVP) for your mobile app, then hybrid can be the option as performance may not be hampered much with less frequent Web-to-native layer interactions. Also, it gives opportunities to the developers to try out different ideas in no time — in the form of MVP of different hybrid mobile apps. They can later opt for the native app route for ideas that gain traction and, hence, need to be extended and scaled in the form of a mobile app (the way Facebook did).

2017: Android and iOS lead the mobile market

With great technological advances continuously enhancing the hardware, both Android and iOS have become the most popular choice of consumers today. With a growing user base, developer adoption and increasing app volume on Play/App Store, Android and iOS are now leading the mobile market with a major share of the global market.

As opposed to BlackBerry, both Android and iOS were launched with their initial focus on the end user. Gradually, the latter two also built security and APIs to cater to business and enterprise needs. With these platforms leading the market and with the growing BYOD culture, both these platforms have become the choice for business apps as well.

Ease of development via an easy and great API suite have resulted in developers and companies quickly adopting these two platforms. As a result, all other mobile platforms/OSs like J2ME, RIM, and even Windows Phone, which came into the market late, have lost their limited market share and almost vanished. This has solved the cost and time-to-market problem caused by supporting multiple mobile OSs/platforms. Developers/companies can now just create apps for Android and iOS. Moreover, with great and simple design wizards to create UIs on Android and with the introduction of simpler programming languages like Swift on iOS, it is now quick and easy to create iOS and Android native apps.

The challenge of apps flooding mobile devices: Developers have started creating separate Android and iOS mobile applications for every small function and are inducing end users to install many of them. This is now creating a problem with users' mobile devices getting flooded with too many applications.

The future: Bot platforms and specialised mobile apps

Today, most mobile users are always connected to social messenger channels like WhatsApp and Facebook Messenger or to mobile OS specific messenger channels like iMessage. This increasing user engagement with messenger channels has influenced messenger app companies like Facebook and Slack to innovate and create bot platforms, which enable companies to create business-specific chat-bots to connect with their end customers on social messenger channels. Apple's iMessage and Google's Allo are both expected to have their bot platforms to help create chat-bots.

Bot platforms are allowing companies to mark their presence on channels where users are investing most of their time. This increases their chances of business. Chat-bots appear as normal messenger contacts to the end user in a messenger list, and thus do away with the need to have separate mobile apps for every small need. Moreover, bot platforms are device and platform neutral — for example, Facebook Messenger runs on all devices, be it mobile phones or wearables, for both Android and iOS.

Bot platforms are new, and deficient in terms of feature support and built-in device capability. So, for the next few years, chat-bots are expected to be used for basic functions and needs instead of having separate mobile apps. Native Android and iOS mobile applications are expected to serve specialised needs/requirements. As an example, it makes more sense to create high performing native mobile banking app for Android and iOS that leverage a built-in fingerprint reader to authorise app access, or use a GPS receiver to identify the location of a user's mobile device and the server list of nearby banks/ATMs. END

By: Gaurav Vohra

The author is a senior solution architect at Impetus Technologies and has more than 13 years of experience in the IT industry. He has been predominantly handling software products and services in the enterprise mobility & digital marketing space. He can be contacted at *gaurav.vohra@impetus.co.in.*